Computer Systems Modelling: Lent 2022 **Assessment 1**
Instructor: Prof. S. Keshav 10-11 AM, February 23, 2023
Answer all questions Maximum marks 15

Question 1. You wish to use simulations to study the mean delay for customer requests waiting in a buffer to be processed by a video server system. The system examines the customer ID and forwards the request either to a high-priority server (for paying customers) or a low-priority server (for non-paying customers). Once a request enters one of these servers, it is never transferred to the other server. Each server has a single finite queue to buffer requests and it serves requests in FIFO order.

a. How would you represent the state of the system? Why? (0.5 marks)

b. Is the system ergodic or non-ergodic? Why? (1 mark)

c. Relying on the previous answer, sketch a simulation approach you would take to simulate the system. (1 mark)

d. State a stopping rule to ensure that, with 99% confidence, the estimate of the mean waiting time is accurate to ±2 ms? (1 mark) Hint: For the 99% confidence interval $z_{\alpha/2} = 2.58$.

e. Discuss how you might use antithetic variables to reduce the simulation time. (0.5 marks)

(Total: 4 marks)

Solution:

a. As a tuple (x,y) where x and y represent the occupancy in the low/high priority buffer.

b. The system is ergodic, since the system is finite, fully connected, and aperiodic.

c. Since it is ergodic, a single run with means computed for each sub-interval suffices.

d. Let n denote the number of sub-intevals, X_i denote the mean waiting time for the ith run, and $\bar{X_n} = 1/n \sum X_i$. First, run the simulation for a certain number n of sub-intervals so that n is not too small. After each subsequent sub-interval, update n and X_i and recompute $\bar{X_n}$. Then, compute $S_n = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X_n})^2}{n-1}}$. Stop if $2.58 * \frac{S_n}{\sqrt{n}} < 2$ ms.

e. Generate request arrivals randomly building on a set of uniform random variables $\{u_i\}$ in even-numbered sub-intervals, using $\{1 - u_i\}$ for odd-numbered sub-intervals.

Question 2. The mean departure rate from a stable, infinite queueing system is 10 customers/second, with inter-departure times drawn from a one-side Gaussian distribution (i.e., all inter-departure times are non-negative). If the mean number of customers in the system is 20, what is a tight lower bound on the mean waiting time?**(2 marks)**

Solution: The system is stable, so the arrival rate λ must be lower than the departure rate $\mu = 10$ customers/s. From Little's law, we know that $20 = \lambda * T$, where T is the mean waiting time. Since $\lambda < \mu = 10$, $20 < 10\ T$, so that $T > 2$s.

Question 3. A state is called absorbing if the probability of leaving this state is 0.

a. Is a Markov chain with an absorbing state irreducible? (0.5 mark)

b. Can a Markov chain with an absorbing state be ergodic? Why or why not? (1 mark)

Total marks for this question: 1.5

Solution:

a. Since we cannot go from an absorbing state to any other state, the chain is not irreducible.

b. Since return to any non-absorbing state is not certain, these states are non-recurrent (transient), and the chain is therefore non-ergodic. The only exception is a degenerate chain with a single absorbing state. (Potential question): what is the only ergodic Markov chain that has an absorbing state.

Question 4. We model the behaviour of a web surfer going from page to page entirely within Wikipedia as a homogeneous, discrete space, discrete time Markov process. We observe that the mean time spent at any page is 10 time units. What is the probability that the surfer leaves the current page at any time step? **(1 mark)**

Solution: The probability that the surfer stays on page i is p_{ii} and that they leave is 1-p_{ii}. Since the mean residence time in a state is $1/(1\text{-}p_{ii}) = 10$, the probability that they leave at any time step is 0.1.

Question 5. Prove that more than one ergodic Markov chain can have the same stationary probability distribution.**(1 mark)**

Solution: By example: two chains, each with three states such that the transition probabilities are symmetric, so that the stationary probability of being in any state is 1/3, but the probabilities differ.

Question 6. The probability that there can be zero arrivals to an M/M/1 queue over 10 seconds is 0.25. What is the probability of 3 arrivals in the same time period? (A final numeric answer is not necessary: an expression will suffice.) **(1 mark)**

Solution: Arrivals follow a Poison process, so we have $e^{-10\lambda} = .25$ so that $\lambda = \frac{ln.25}{-10} s^{-1}$. The probability of 3 arrivals is $\frac{(-10\lambda)^3}{3!} e^{-10\lambda} = 0.125 \text{ ln } .25$.

Question 7. In effort to be responsive yet efficient, a supermarket manager introduces the following rule: normally there will be only one checkout counter open. However, if five customers are waiting for service (including the customer being served), then every customer would get their own checkout counter. The supermarket reverts to a single line whenever the total number of customers waiting for service or in service drops below five.

a. How would you model this system? **Clearly state all assumptions**. (1 mark)

b. What is the longest time someone would need to wait for service (1.5 marks)

Total marks for this question: 2.5

Solution:

a. Assume that customers arrive as a Poisson process and that service durations are exponential. Also assume that it takes negligible time for customers to go from one queue to many. Then, the system can be modeled as an M/M/1 queue as long as the total number of customers waiting for service (including when in line) is less than five, and as an M/M/∞ queue otherwise.

b. There is no bound on the longest wait time, since this is a stochastic process. However, the longest expected waiting time would be for the the fourth customer in the single queue. On average, they would need to wait for 3.5 mean service times, which incurs an expected delay of $3.5/\mu$ time units.

Question 8. Jobs arrive to a server as a Poisson process at a mean rate of 2 jobs/s and are served in FIFO order by a server with a mean service rate of 3 jobs/s. What is the largest

variance tolerable in the server if the mean waiting time (including service time) should be less than 5 seconds? **(2 marks)**

Solution: The mean service rate $\mu = 3$ and the mean arrival rate $= \lambda = 2$. So, $\rho = \lambda/\mu = 2/3$. The mean service time is $1/\mu = 1/3$ s. C_S^2, the square of the coefficient of variation, is defined as the ratio of the variance to the square of the mean service time. Let V denote the variance. Then, $C_S^2 = V/(1/3)^2 = 9V$. The expected sum of the time waiting for service and the service time, is given by:

$$\mathbb{E}(T) = \frac{1}{\mu}(1 + \frac{\rho(1 + C_s^2)}{2(1 - \rho)}) \tag{1}$$

Plugging in the values:

$$\mathbb{E}(T) = \frac{1}{3}(1 + \frac{2/3(1 + 9V)}{2(1 - 1/3)}) \tag{2}$$

$$\mathbb{E}(T) = \frac{1}{3}(1 + \frac{2/3(1 + 9V)}{2/3}) \tag{3}$$

$$\mathbb{E}(T) = \frac{1}{3}(1 + 1 + 9V) \tag{4}$$

We want this quantity to be less than 5, and after simplification, we get $V < 13/9$.